4d. For-Loops

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INTRODUCTION

A key feature of programming is its ability to automate repetitive tasks, rendering **for-loops** crucial in coding. They let you execute the same block of code multiple times, treating each element in a list as a different input.

Although all languages provide some form of looping, their performance characteristics and idiomatic use vary widely across them. For instance, many high-level languages designed for scientific computing (e.g., MATLAB, Python, and R), programmers are typically encouraged to avoid explicit for-loops when performance matters. Instead, these languages favor vectorized operations (which apply a single operation to an entire array at once) or specialized library calls (which use pre-optimized routines). The design reflects a practical reality: for-loops in those languages are often slow, as their internal design lacks the necessary mechanisms to optimize them automatically.

In contrast, for-loops in Julia aren't only idiomatic but also fast, often matching or surpassing the performance of vectorized alternatives. As a result, mastering for-loops isn't just a matter of stylistic preference. Rather, it's an essential skill for writing efficient programs in Julia.

Part II of this book will examine how for-loops contribute to high performance. For now, our focus is on understanding the construct itself: its syntax, its variations, and the most common patterns for iterating over data structures.

SYNTAX

For-loops delimit their scope via the keywords `for` and `end`. To illustrate the syntax, consider the function `println(a)`, which evaluates `a` and displays its output in the REPL. In case `a` is a string, `println(a)` simply displays the word stored in `a`. The following script repeatedly applies `println` to display each word contained in a collection.

FOR-LOOP SYNTAX

```
for x in ["hello", "beautiful", "world"]
    println(x)
end
```

```
hello
beautiful
world
```

Remark

The keyword `in` can be replaced by `∈` or `=`, where `∈` can be written through tab completion using the command `\in`. Considering this, the following constructions are all equivalent.

IN

```
for x in ["hello", "beautiful", "world"]
    println(x)
end
```

∈

```
for x ∈ ["hello", "beautiful", "world"]
    println(x)
end
```

=

```
for x = ["hello", "beautiful", "world"]
    println(x)
end
```

Furthermore, we can employ any character or term to describe the iteration variable. For instance, we iterate below using `word`.

ALTERNATIVE NAME FOR ITERATION VARIABLE

```
for word in ["hello", "beautiful", "world"]
    println(word)
end
```

```
hello
beautiful
world
```

Based on this example, we can identify three components that characterize a for-loop:

- **A code block to be executed**: represented in the example by `println(x)`, which shows the value of `x` in the REPL.
- **A list of elements**: represented in the example by `["hello","beautiful","world"]`. This specifies the elements over which we'll apply the code block. The list can contain elements with any data type (e.g., strings, numbers, and even functions). The only requirement is that the list must be an **iterable object**, defined as a collection whose elements can be accessed individually. An example of iterable object is vectors, as in the example. However, we'll also introduce others that are most commonly used, such as ranges.

- **An iteration variable**: represented in the example by `x`. This serves as a label that holds the value of each element in the list during iteration. Iteration variables are local variables, with no significance outside the for-loop. Their sole purpose is to provide a convenient way to access and manipulate the elements of the list.

In the following sections, we'll explore different collections that are iterable and therefore can serve as lists. Furthermore, we'll show that these lists can comprise elements not immediately obvious. A typical example is functions, making it possible to apply different functions to the same object.

> **Always Wrap For-Loops in Functions**
>
> At this stage, we're still introducing fundamental concepts. Thus, we're presenting subjects in their simplest form for learning purposes. In particular, this explains why for-loops will be primarily written in the global scope.
>
> However, **you should always wrap for-loops in functions**. Executing for-loops outside a function severely degrades performance, and is additionally subject to different rules regarding variable scope. [1]

ITERATING OVER INDICES

So far, we've considered a simple list like `["hello", "beautiful", "world"]` to demonstrate how for-loops work. In real applications, however, manually specifying each element in a collection is impractical. Fortunately, when a list follows a predictable pattern (e.g., a sequence of numbers), we can simply describe the pattern that generates those elements.

Building on this insight, we'll next explore how to define ranges. They let users define a sequence of numbers, which is particularly useful to access elements of a collection through their indices.

RANGES

Ranges in Julia are defined via the syntax `<begin>:<steps>:<end>`, where `<begin>` represents the starting index and `<end>` the ending index. Likewise, `<steps>` sets the increment between values, defaulting to one when the term is omitted. We can also reverse the order of the sequence, by providing a negative value for `<steps>`. All this is demonstrated below.

RANGE WITH STEPS GIVEN

```
for i in 1:2:5
    println(i)
end
```

```
1
3
5
```

RANGE WITH REVERSE ORDER

```
for i in 3:-1:1
    println(i)
end
```

```
3
2
1
```

> **Remark**
>
> The application of ranges isn't limited to for-loops. They can also be used to define vectors by combining them with the `collect` function.
>
> **CREATING A VECTOR FROM A RANGE**
>
> ```
> x = collect(4:6)
> ```
>
> ```
> 3-element Vector{Int64}:
> 4
> 5
> 6
> ```

ITERATING OVER INDICES OF AN ARRAY

Ranges provide a straightforward mechanism for traversing the elements of a collection. When used in combination with a for-loop, they let you access each element of a vector by its index. There are several ways to construct such ranges.

A straightforward method is to write `1:length(x)`, which generates all valid indices for the vector `x`. Since `length(x)` returns the number of elements, this range covers every position from the first to the last. While this approach works, it relies on the assumption of linear indexing starting at 1, which isn't always guaranteed. As a result, it can be fragile when applied to collections with different indexing schemes.

A more robust practice is to use `eachindex(x)`. This function produces an iterator that's optimized to the specific structure of the collection. Furthermore, it ensures that you're iterating over the correct set of indices, regardless of the underlying data type. This makes your code more general, more

efficient, and less error-prone, especially when working with other iterable objects that may not use standard indexing.

1:LENGTH(X)

```
x = [4, 6, 8]

for i in 1:length(x)
    println(x[i])
end
```

```
4
6
8
```

EACHINDEX

```
x = [4, 6, 8]

for i in eachindex(x)
    println(x[i])
end
```

```
4
6
8
```

Julia provides other methods to iterate over all indices of a collection. Two constructions worth mentioning are `LinearIndices(x)` and `firstindex(x):lastindex(x)`. Just like the previous methods defined, they create a range that goes from the first to the last index of `x`.

All these approaches can be used interchangeably, as shown below.

EACHINDEX

```
x = [4, 6, 8]

for i in eachindex(x)
    println(x[i])
end
```

```
4
6
8
```

1:LENGTH(X)

```
x = [4, 6, 8]

for i in 1:length(x)
    println(x[i])
end
```

```
4
6
8
```

LINEARINDICES

```
x = [4, 6, 8]

for i in LinearIndices(x)
    println(x[i])
end
```

```
4
6
8
```

FIRSTINDEX(X):LASTINDEX(X)

```
x = [4, 6, 8]

for i in firstindex(x):lastindex(x)
    println(x[i])
end
```

```
4
6
8
```

The multiplicity of methods to implement the same functionality is necessary to handle non-standard indices. For instance, the `OffsetArrays.jl` package sets the first index of arrays to 0, a common convention in many programming languages. When this package is loaded, using `1:length(x)` would break portability, while `firstindex(x):lastindex(x)` adapts seamlessly to the modified indexing scheme.

Unless you're developing a package for other users, you don't need to worry much about which approach to implement.

VARIABLE SCOPE IN FOR-LOOPS

For-loops introduce a new variable scope, which is largely governed by the same rules as functions. Specifically, given a variable named `x`, the rules of variable scope are as follows:

- If `x` is the variable of iteration, then `x` is always local to the for-loop. This holds regardless of whether there's a variable `x` defined outside the for-loop.
- If there's a variable named `x` outside the for-loop and no `x` is defined within the for-loop, `x` refers to the global variable. Moreover, the for-loop can mutate the value of `x`.
- If there's a variable `x` defined within the for-loop but no `x` defined outside it, `x` is local and won't be accessible outside the for-loop.

The following code snippets illustrate these rules. The third example is particularly noteworthy, as it highlights **a common beginner mistake**: defining a variable inside a for-loop and then attempting to use it outside the for-loop, only to discover that it's no longer accessible.

ITERATION VARIABLES ARE LOCAL

```
x = 2

for x in ["hello"]          # this 'x' is local, not related to 'x = 2'
    println(x)
end
```

```
hello
```

NEW VARIABLES ARE LOCAL

```
#no `x` defined outside the for-loop

for word in ["hello"]
    x = word                # `x` is local to the for-loop, not available outside it
end
```

```
julia> x
ERROR: UndefVarError: x not defined
```

X REFERS TO GLOBAL X

```
x = [2, 4, 6]

for i in eachindex(x)
    x[i] * 10              # refers to the `x` outside of the for-loop
end
```

```
julia> x
3-element Vector{Int64}:
 2
 4
 6
```

GLOBAL X MUTATED

```
x = [2, 4, 6]

for i in eachindex(x)
    x[i] = x[i] * 10        # mutates the `x` defined outside the for-loop
end
```

```
julia> x
3-element Vector{Int64}:
 20
 40
 60
```

> ⚠ **Warning!**
>
> The scope rule that we'll present next always applies, except in one edge case. Since this case primarily results from discouraged coding practices, we briefly outline it without further elaboration.
>
> Specifically, the rule won't apply when three conditions occur simultaneously: *i)* the for-loop is *not* inside a function, *ii)* a local variable shares the same name as a global variable, and iii) the script is run non-interactively (i.e., executing code through a script file). [2] When these conditions are met, Julia prevents reassignment of variables inside for-loops.
>
> **You can safely ignore this exception** for two reasons. First, due to performance reasons that we'll explore later in this book, you should always write for-loops within functions. Second, even if this scenario occurs, Julia will issue a warning in the REPL, alerting you that the code may not behave as intended.

The last scope rule applies when for-loops appear inside functions or when code is run interactively:

- if there's a variable `x` defined both within the for-loop and outside it, `x` will be reassigned to the value given inside the for-loop.

The following example demonstrates this rule, where we enclose the for-loop in a function.

REASSIGNMENT OF X

```
function foo()
    x = [2, 4, 6]

    for word in ["hello"]
        x = word              # `x` is reassigned
    end

    return x
end
```

```
julia> foo()
"hello"
```

ARRAY COMPREHENSIONS

Julia provides **array comprehensions** as a concise and expressive way to generate new arrays via for-loops. The general syntax is `[<expression> for... if...]`, where `<expression>` denotes either an operation or a function.

To illustrate, suppose we want to define a new vector `y`, whose elements are the squares of the corresponding entries in `x`. Array comprehensions make this task straightforward. The following code snippets demonstrate the approach by using a direct expression and by calling a function.

ARRAY COMPREHENSION WITH OPERATION

```
x      = [1,2,3]


y      = [a^2 for a in x]
z      = [x[i]^2 for i in eachindex(x)]
```

```
julia> y
3-element Vector{Int64}:
 1
 4
 9
julia> z
3-element Vector{Int64}:
 1
 4
 9
```

ARRAY COMPREHENSION WITH FUNCTION CALL

```
x      = [1,2,3]
foo(a) = a^2

y      = [foo(a) for a in x]
z      = [foo(x[i]) for i in eachindex(x)]
```

```
julia> y
3-element Vector{Int64}:
 1
 4
 9
julia> z
3-element Vector{Int64}:
 1
 4
 9
```

Array comprehensions can also incorporate conditions, allowing you to filter elements as they're generated. In such cases, the condition must be placed at the end of the comprehension.

ARRAY COMPREHENSION THAT ADDS CONDITION

```
x = [1, 2, 3, 4]

y = [a for a in x if a ≤ 2]
z = [x[i] for i in eachindex(x) if x[i] ≤ 2]
```

```
julia> y
2-element Vector{Int64}:
 1
 2
julia> z
2-element Vector{Int64}:
 1
 2
```

Array Comprehensions for Defining Matrices

Array comprehensions can also be used to construct matrices. In this case, the syntax requires a comma to separate the iteration variables for each dimension.

ARRAY COMPREHENSION FOR MATRICES

```
X = [i * j for i in 1:2, j in 1:2]
```

```
julia> X
2×2 Matrix{Int64}:
 1  2
 2  4
```

ITERATING OVER MULTIPLE OBJECTS

For-loops can handle more than just single-value iterations, also supporting **simultaneous iteration over multiple values**.

We'll focus on two common scenarios: simultaneously iterating over two iterable collections, and iterating over both the indices and values of a single collection. In each case, we'll explore how to implement the iteration using both plain for-loops and array comprehensions, highlighting the differences in syntax and use cases.

SIMULTANEOUSLY ITERATING OVER TWO COLLECTIONS

Consider two iterable objects `x` and `y`. There are two strategies for combining their elements, depending on the desired outcome.

First, the function `Iterators.product(x,y)` generates all the possible combinations of elements from `x` and `y`. Thus, it produces a pair `(x[i], y[j])`, where `i` and `j` take all possible indices. The `Iterators.product` function is part of the package `Iterators`, imported by default in each Julia session.

Alternatively, it's possible to iterate over all ordered pairs of `x` and `y`. This is implemented through the function `zip(x,y)`, which provides the pair of *i*-th elements from `x` and `y` in the *i*-th iteration. Thus, compared to `Iterators.product`, it restricts the iterations to the pairs `(x[i], y[i])`.

PAIRS

```
list1 = ["A","B"]
list2 = [ 1 , 2 ]

for (a,b) in zip(list1,list2)                    #it takes pairs with the same index
    println((a,b))
end
```

```
("A", 1)
("B", 2)
```

ALL COMBINATIONS

```
list1 = ["A","B"]
list2 = [ 1 , 2 ]

for (a,b) in Iterators.product(list1,list2)    #it takes all possible combinations
    println((a,b))
end
```

```
("A", 1)
("B", 1)
("A", 2)
("B", 2)
```

SIMULTANEOUSLY ITERATING OVER INDICES AND VALUES

When iterating over a collection, there are scenarios where you need access to both the value and index of an element, rather than handling each separately. For example, when analyzing data, you may want to flag outliers not only by their magnitude, but also by their position in the dataset.

Julia provides a direct way to achieve this through the `enumerate` function. This transforms a collection into an iterator that yields pairs of indices and values, thus providing access to both pieces of information during each iteration.

FOR-LOOPS

```
x = ["hello", "world"]

for (index,value) in enumerate(x)
    println("$index $value")
end
```

```
1 hello
2 world
```

ARRAY COMPREHENSION

```
x = [2, 3]


y = [(x[index] - value) for (index,value) in enumerate(x)]    # `x[index]` equals `value`
```

```
2-element Vector{Int64}:
 0
 0
```

ITERATING OBJECTS TO CREATE VECTORS

To generate vectors that iterate over multiple values, we can employ array comprehension. In particular, to construct a vector that exhausts all combinations between two iterable collections, Julia offers a special syntax: simply repeat the `for` keyword to specify the iteration range of each variable.

Note that the `for` clauses must be written sequentially, without commas. A comma changes the comprehension's behavior, instructing Julia to build a matrix instead of a vector, as demonstrated before.

We illustrate these cases, along with the vectors obtained with `zip` and `Iterators.product` for this purpose.

ALL COMBINATIONS

```
list1 = ["A","B"]
list2 = [ 1 , 2 ]

x     = [(i,j) for i in list1 for j in list2]
```

```
julia> x
4-element Vector{Tuple{String, Int64}}:
 ("A", 1)
 ("A", 2)
 ("B", 1)
 ("B", 2)
```

NOTATION FOR MATRICES

```
list1 = ["A","B"]
list2 = [ 1 , 2 ]

X     = [(i,j) for i in list1, j in list2]                    # this defines a matrix
```

```
julia> x
2×2 Matrix{Tuple{String, Int64}}:
 ("A", 1)  ("A", 2)
 ("B", 1)  ("B", 2)
```

ZIP

```
list1 = ["A","B"]
list2 = [ 1 , 2 ]

x     = [(i,j) for (i,j) in zip(list1,list2)]
```

```
julia> x
2-element Vector{Tuple{String, Int64}}:
 ("A", 1)
 ("B", 2)
```

ITERATORS.PRODUCT

```
list1 = ["A","B"]
list2 = [ 1 , 2 ]

X     = [(i,j) for (i,j) in Iterators.product(list1,list2)]    # this defines a matrix
```

```
julia> X
2×2 Matrix{Tuple{String, Int64}}:
 ("A", 1)  ("A", 2)
 ("B", 1)  ("B", 2)
```

ITERATING OVER FUNCTIONS

Functions in Julia are **first-class objects**, also referred to as **first-class citizens**. This means that functions can be manipulated just like any other data type, such as strings and numbers. In particular, this property makes it possible to store functions in a vector and apply them sequentially to an object. The following example illustrates this by computing descriptive statistics of a vector `x`.

ITERATION OVER FUNCTIONS

```
x                       = [0, 5, 10]
list_functions          = [minimum, maximum]

descriptive(vector,list) = [foo(vector) for foo in list]
```

```
julia> descriptive(x, list_functions)
2-element Vector{Int64}:
  0
 10
```

FOOTNOTES

[1] In fact, older versions of Julia were restricting the use of for-loops in the global scope.

[2] There are two methods to run code. The first method is the one used thus far, where we work interactively with Julia. This includes both running commands in the REPL and executing lines through a code editor. The second method consists of executing files containing scripts, which is done with the function `include`.